

ASSIGNMENT-02.

NAME : DINDU. PUJITHA.

REG NO : 192373037.

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Set 1:

DES problem: Given a plaintext block and a simplified DES key, perform one round of DES and show all intermediate values.

→ DES uses a 64-bit plaintext block and a 64-bit key.

→ one round involves:

- * Initial permutations (IP).
- * Split into L_0 and R_0
- * Expand R_0 from 32 → 48 bits.
- * XOR with round key.
- * S-Box substitution.
- * P-Box permutation.
- * XOR with L_0

→ Result:

$$L_1 = R_0$$

$$R_1 = L_0 \oplus F(R_0, K_1)$$

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compare symmetric and asymmetric encryption using real-time applications (e.g., WhatsApp vs. Email encryption).

symmetric	Asymmetric
uses one key	uses public+private keys.
Faster	Slower
Example: AES	Example: RSA
used for bulk data	used for key exchange / authentication.
WhatsApp uses symmetric encryption for messages.	Email systems may use asymmetric encryption for secure key exchange.

3. RSA problem. choose two small primes $p=7$, $q=11$. Generate

RSA key and Encrypt the message $M=9$.

Given:

$$p=7, q=11, M=9$$

Step 1:

$$n = p \times q = 7 \times 11 = 77$$

Step 2:

$$\phi(n) = (7-1)(11-1) = 60$$

choose $e=7$, since $\gcd(7, 60)=1$

Step 3:

find d :

$$7 \times d = 1 \pmod{60}$$

$$d=43$$

public key: $(7, 77)$

private key: $(43, 77)$

Encryption:

$$C = M^e \pmod{n}$$

$$C = 9^7 \pmod{77} = 37$$

Ciphertext = 37

4. Explain CBC and CTR modes of operation with block diagrams. which one is more secure for parallel processing?

CBC (Cipher Block Chaining):-

plaintext $\rightarrow$ XOR $\leftarrow$ IV

$\downarrow$
Encrypt

$\downarrow$
ciphertext

$\downarrow$
Next Block.

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- * Each block depends on the previous ciphertext.

- * Encryption is not easily parallelized.

CTR (Counter Mode) :-

counter $\rightarrow$ Encrypt $\rightarrow$ keystream

plaintext $\xrightarrow{\text{XOR}}$ ciphertext

- * Each block uses a different counter.

- * Encryption/decryption can be parallelized.

CTR is better for parallel processing because blocks can be processed independently.

5. Define Quantum cryptography. How does quantum key distribution (QKD) solve the key distribution problem?

Quantum cryptography:

Uses principles of quantum mechanics to provide secure communication.

QKD (Quantum Key Distribution) :-

- * Allows two users to securely generate a shared secret key.

- * If an attacker tries to observe the quantum particles, their state changes.

- * This reveals the presence of an attacker.

Example:- BB84 protocol is a popular QKD protocol.

Main advantage:- It provides secure key distribution and detects eavesdropping.

Set 2 :-

1. Blowfish problem: Given an 8-bit input value 0110100101001011001001 (decimal 105), show how S-box 1 is used in the F-function.

Given input = 105 = 01101001

* Divide into 8-bit value: 01101001

* Decimal value = 105

* S-box 1 is accessed using this value as the index:

$S_1[105]$

* The corresponding S-box value is then used in the F-function.

Note :-

A complete Blowfish F-function normally uses a 32-bit input, so the exact S_1 output cannot be calculated from only one 8-bit value without the specific S-box table.

2. Describe the Diffie-Hellman Key Exchange and compute a shared key with:

* public Base $g = 5$, prime $P = 23$.

* Alice's private key = 6, Bob's private key = 15.

Given:

$g = 5, P = 23$

Alice private key = 6

Bob private key = 15

Alice public key: $A = 5^6 \bmod 23 = 8$

Bob public key: $B = 5^{15} \bmod 23 = 19$

shared key: $K = 19^6 \bmod 23 = 2$

shared key = 2

compare Blowfish, RC5, and DES in terms of key size, structure, and performance.

Feature	Blowfish	RC5	DES
Key size	32-448 bits	0-2048 bits	56 bits
Structure	Feistel	Feistel-like	Feistel
Block size	64 bits	variable	64 bits
performance	fast	fast	slow
security	strong	strong	weak today.

4. Distinguish b/w block and stream ciphers. provide an example scenario for each.

Block cipher :-

- * Encrypts fixed-size blocks.
- * Example : AES.
- * Scenario : Encryption files.

Stream cipher :-

- * Encrypts data bit-by-bit or byte-by-byte.
- * Example : chacha20.
- * Scenario : Real-time communication / streaming.

5. why is key management critical in public key crypto systems? Explain with digital certificate usage.

Key management is important to:

- * protect private keys.
- * verify public keys.
- * prevent impersonation.

Digital certificate:-

A certificate binds a public key to an identity and is issued by a trusted certificate authority (CA).

Example:-

HTTPS websites use digital certificates to prove their identity.

Set

Set 3:-

Describe the full process of Triple DES (3DES)

Encryption using two keys: $K_1 = 101010$, $K_2 = 010101$, on data block $D = 110011$.

3DES uses Encrypt $\rightarrow$ Decrypt $\rightarrow$ Encrypt (EDE).

Given:

$$K_1 = 101010$$

$$K_2 = 010101$$

$$D = 110011$$

Process:

Step 1: Encrypt D using K_1

$$E(K_1, D)$$

Step 2: Decrypt the result using K_2

$$D(K_2, E(K_1, D))$$

Step 3: Encrypt again using K_1

$$E(K_1, D(K_2, E(K_1, D)))$$

Final ciphertext = $E(K_1, D(K_2, E(K_1, D)))$.

4

What is a meet-in-the-middle attack? Illustrate how it applies to breaking double DES.

A meet-in-the-middle (MITM) attack is used against double DES.

Double DES:-

$$C = E(K_2, E(K_1, P))$$

Attacker:-

1. Encrypt plaintext P with all possible K_1 .
2. Decrypt ciphertext C with all possible K_2 .
3. Find matching intermediate values.
4. Matching pair gives K_1 and K_2 .

It reduces the effective security of double DES.

3. RSA Problem: Encrypt and decrypt message $M=5$ using $P=3$, $q=11$, $e=3$.

Given:

$$P=3, q=11, M=5, e=3$$

$$n = p \times q = 33$$

$$\phi(n) = (3-1)(11-1) = 20$$

Find d :

$$3d \equiv 1 \pmod{20}$$

$$\text{So, } d=7$$

$$\text{Public Key} = (3, 33)$$

$$\text{Private Key} = (7, 33)$$

Encryption:-

$$C = 5^3 \bmod 33 = 26$$

Decryption:-

$$M = 26^7 \bmod 33 = 5$$

Encrypted message = 26

Decrypted message = 5

4.

Define padding in block ciphers. Show how padding would be added to the message "HELLO" in 8-byte blocks.

padding adds extra bytes so the message becomes exactly the required block size.

Message:

HELLO = 5 bytes.

Block size = 8 bytes.

Using PKCS#7 padding:

HELLO 03 03 03

so the 8-byte block is:

HELLO 03 03 03.

5

Discuss the use of elliptic curve cryptography (ECC) for secure messaging in IoT devices.

ECC (Elliptic curve cryptography) provides strong security with small keys.

For IoT devices, ECC is useful because:

- * Requires less memory.
- * Uses less power.
- * Provides fast encryption/key exchange.
- * Suitable for resource-constrained devices.

Example: IoT sensor ↔ Gateway can use ECC for secure key exchange and encrypted messaging.

Set 4 :-

1. Feistel problem: perform one round of a Feistel structure with input $L=1010$, $R=1100$, subkey $K=1111$, and $F(R, K) = (R \text{ XOR } K)$.

Given :-

$$L = 1010$$

$$R = 1100$$

$$K = 1111$$

$$F(R, K) = R \text{ XOR } K$$

$$1100 \text{ XOR } 1111 = 0011$$

Feistel Equations:-

$$L_1 = R_0 = 1100$$

$$R_1 = L_0 \text{ XOR } F(R_0, K)$$

$$1010 \text{ XOR } 0011 = 1001$$

output :-

$$L_1 = 1100, R_1 = 1001$$

2. Explain how RSA and Diffie-Hellman can be used together in a secure communication session (e.g., hybrid cryptosystem).

Diffie-Hellman :

Establishes a shared secret key

RSA: Authenticates users/signs messages.
 * The shared key is then used with AES for fast data encryption.

Flow:-

DH $\rightarrow$ Shared Key $\rightarrow$ AES Encryption.

RSA $\rightarrow$ Authentication.

3. compare the s-box operation in Blowfish and DES.
 why is s-box non-linearity important?

Blowfish	DES.
16 s-boxes	8 s-boxes
Each has 256 entries	Each has 64 entries
32-bit input	6-bit input.
more complex	less complex.

Non-linearity makes it difficult for attackers to find mathematical relationships between plaintext, key, and ciphertext.

4. Distinguish between ECB, CBC, OFB and CTR modes in terms of error propagation and parallelism.

Mode	Error propagation	parallelism
ECB	only affected block	yes.

CBC	Affects current & next block	Encryption no limited.
OFB	Bit error affects same bit	yes.
CTR	only affected bits	

Best for parallel processing : CTR and ECB.

5. What is a man-in-the-middle attack? propose a method to defend against it in a public key system.

A MITM attack occurs when an attacker secretly intercepts communication b/w two users.

Defense:

- * Use digital certificates.

- * Verify public key using a trusted certificate authority (CA).

- * Use authenticated key exchange such as TLS.

This prevents attackers from substituting their own public key.

Set 5:-

1. Describe in detail the Blowfish Encryption steps, including key expansion, subkeys, and F-function processing.

Blowfish is a symmetric block cipher with a 64-bit block size and variable key size (32-448 bits).

Main steps:-

Key expansion:

The user key is expanded into subkeys.

P-array :
Generates 18 subkeys : P_1 to P_{18} .

S-boxes :
uses 4 S-boxes, each containing 256 entries.

F-function :

A 32-bit inputs is divided into four 8-bit parts and processed through the S-boxes.

16 Feistel rounds;

Each round uses a P-array subkey and the F-function.

Final step :

The two 32-bit halves are combined to produce the 64-bit ciphertext.

2.

Diffie-Hellman problem: If $P=17$, $g=3$, and private keys are $a=6$ and $b=13$, compute the public keys and shared secret.

Given :

$$P=17, g=3$$

$$\text{Alice : } a=6$$

$$\text{Bob : } b=13$$

Alice public key :

$$A = 3^6 \bmod 17 = 15$$

Bob public key :

$$B = 3^{13} \bmod 17 = 12$$

Shared secret :

$$K = B^a \bmod 17 = 12^6 \bmod 17 = 4$$

public keys : $A=15$, $B=12$

shared secret = 4

compare the security and efficiency of DES, 3DES, RSA and ECC. In what scenarios would you use each?

Algorithm	security	Efficiency	use
DES	weak	fast	legacy systems.
3DES	Better than DES, but outdated	slow	legacy financial systems.
RSA	strong with large keys	slow	Digital signatures, key exchange.
ECC	Strong	Fast & small keys	Mobile, IoT, modern secure communication.

Best modern choice :-

ECC for many public-key applications; symmetric ciphers are generally used for bulk data.

4. Define and differentiate ~~block~~ stream and block ciphers with real-time use cases (eg, video streaming vs. file Encryption).

stream cipher :-

- * Encrypts data continuously, bit-by-bit or byte-by-byte.
- * Fast and suitable for real-time data.

* Example:-

video/audio streaming using chacha20.

Block cipher:-

* Encrypts fixed size blocks.

* Suitable for stored data.

* Example:-

File encryption using AES.

5. Explain how quantum computing can break RSA and how post-quantum cryptography is being developed to address this.

Quantum computing and RSA:-

* RSA security depends on the difficulty of factoring large numbers.

* A sufficiently powerful quantum computer can use Shor's algorithm to factor these numbers efficiently.

* Therefore, RSA could become insecure against large-scale quantum computers.

Post-quantum cryptography (PQC):-

* Develops algorithms that are resistant to quantum attacks.

* Examples include lattice-based, hash-based, and code-based cryptography.

* These algorithms are being standardised and deployed to prepare for the quantum era.

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